

Structure over Scale: Hierarchy-Constrained Modeling for English Online Polarization Detection

Mid-Submission Progress Report

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Abstract

In SemEval-2026 Task 9, a text is polarized exactly when at least one social dimension applies, yet shared-task systems predict the two subtasks separately and repair contradictions after decoding. We report the first stage of a controlled study of this hierarchy on English. Five conditions share one DeBERTa-v3-base encoder: independent models (M1), a multi-task model (M2), post-hoc gating of M2 (M3), and M4-core, which derives detection from the dimension probabilities through a differentiable noisy-OR. Over 5 folds $\times$ 5 seeds (100 training runs), with paired bootstrap confidence intervals, the unconstrained models contradict the hierarchy on 62% of all texts, while M3 and M4-core reach zero violations. Under our pre-registered metrics M4-core does *not* improve macro-F1 over M1 or M2 (RQ1): TYPE macro-F1 is -0.082 and -0.036 lower. At equal consistency it is statistically tied with gating (RQ2: $+0.008$, 95% CI $[-0.009, +0.022]$), while recovering more labels on multi-label texts ($+0.025$). A gating audit shows that M3 deletes 23% of the multi-task model’s correct dimension labels, because 98.7% of its detection misses already carried a correct dimension. We trace M4-core’s detection deficit to a decoding rule that tunes dimension thresholds without seeing non-polarized text, and show that the TYPE ranking reverses when all texts are scored.

1 Introduction

Polarized language carries hostility, stereotyping, exclusion, or divisive framing. The signals are contextual rather than lexical: political vocabulary alone does not make a text polarized, and one text can involve several social dimensions at once. SemEval-2026 Task 9 (Naseem et al., 2026) splits the problem into POLARDETECT (DET: is the text polarized?) and POLARTYPE (TYPE: which of Political, Racial/ethnic, Religious, Gender/sexual identity, and Other apply?).

The two subtasks are *nested*, not merely related. TYPE labels exist only for polarized texts, so a text is polarized if and only if some dimension applies. Standard multi-task learning puts independent heads on a shared encoder, so nothing represents this constraint. Guo and Chang (2026) note that independent modelling “causes hierarchical violations”. Maharjan et al. (2026) measure violation rates of up to 33% and remove them with a deterministic rule at inference. They report that the rule “slightly limits raw multi-label recall”: on English, gating lifts DET from 0.752 to 0.826 but lowers TYPE from 0.511 to 0.506.

Every published fix acts on outputs. We ask whether the recall cost belongs to the constraint itself or to the point at which it is applied. Our structured model, M4-core, has no detection head. It computes the detection probability from the five dimension probabilities through a noisy-OR, so contradictory outputs cannot be represented. On non-polarized text, the detection loss reaches the dimension detectors, which is exactly where TYPE supervision is missing.

This report covers the mid-submission scope of the project. It addresses two contributions:

- **C1 / RQ1.** Does deriving DET from TYPE via noisy-OR improve DET and/or TYPE macro-F1 over independent models (M1) and unconstrained multi-task learning (M2)?
- **C2 / RQ2.** At equal hard-prediction consistency, does training-time structure (M4-core) preserve more TYPE recall and F1 than post-hoc deterministic reconciliation (M3)?

Label-aware representations (C3/RQ3) and the parameter-efficiency comparison with 12B–27B systems (C4/RQ4) are scheduled after the mid-submission (§8).

Summary of findings. Unconstrained models violate the hierarchy on 62% of texts, while M3

and M4-core reach 0%. RQ1 is *not supported*: M4-core is below both baselines. RQ2 is *inconclusive*: M4-core ties with gating on macro metrics, but is better on multi-label texts and more stable across seeds. A gating audit quantifies the error propagation behind M3’s recall loss. Two protocol choices shape the RQ1 outcome: TYPE is scored only on polarized texts, and M4-core’s thresholds are tuned without seeing non-polarized text. Both are documented here and are the first targets of the next stage.

2 Background and Related Work

The POLAR benchmark. SemEval-2026 Task 9 covers 22 languages and attracted 67 teams and 73 system papers (Naseem et al., 2026). Two properties of the English data shaped our design. First, annotator agreement is only fair (Fleiss’s $\kappa = 0.39$, against 0.83 for Khmer), so part of the label signal is genuinely contested. Second, headroom differs sharply between subtasks. The best English DET score is 0.825 against a 0.773 baseline, whereas TYPE reaches 0.532 against 0.347. Detection is close to saturated; dimension classification is not.

POLAR systems. Large generative models dominated the task: Qwen was used by 31% of teams, LLaMA by 20%, and Gemma/Gemini by 19%, against about 7% for BERT-style encoders. Both English subtask winners have 12B–27B parameters (Naseem et al., 2026; Tangkulung and Tsuruoka, 2026; Lin et al., 2026). Among encoder systems, Daniyal et al. (2026) compare BERT variants, Tran (2026) add an LLM refinement stage to DeBERTa, Orsanigo et al. (2026) report multi-task gains, and Nestor et al. (2026) use label-wise attention. Our protocol follows Bokaei et al. (2026): cross-validation with out-of-fold threshold calibration.

Hierarchy and consistency. Almost every team tunes per-label thresholds (Naseem et al., 2026), so we treat that as standard practice. Guo and Chang (2026) observe violations but leave their measurement and correction to future work. Lin et al. (2026) check the DET prediction before assigning TYPE labels. Mohammad (2026) route inputs through a binary gatekeeper that zeroes child labels for non-polarized text, and force an argmax when a polarized text fires nothing. Maharjan et al. (2026) formalise the Logical Violation Rate

Statistic (English training split)	Value
Texts	3,222
polarized (DET= 1) / non-polarized	1,175 / 2,047
TYPE support: Political	1,150
Racial/ethnic / Religious	281 / 112
Gender/sexual / Other	72 / 126
Polarized texts with ≥ 2 / exactly 1 TYPE	426 / 749
<i>Other</i> co-occurring with a named label	121
Hierarchy violations in gold labels	0

Table 1: The English training split, on which every result in this report is computed (out-of-fold). Political appears in 98% of polarized texts, so it is almost a proxy for DET. Gender/sexual and Religious are the rare labels.

(LVR), audit it across all languages, enforce $\hat{y}_{S1} = 0 \Rightarrow \hat{y}_{S2} = 0$ at inference, and feed the child task *detached* parent logits. That stop-gradient deliberately prevents the tasks from shaping each other’s representations.

The gap. In every system, the constraint acts on outputs: detection remains an independent prediction that is overruled later. We found no Task 9 system in which detection is parameterised as a function of the dimension predictions. Hierarchy-consistent neural classifiers are established in structured prediction (Giunchiglia and Lukasiewicz, 2020), multi-task learning has a long history (Caruana, 1997; Kendall et al., 2018), and label-wise attention goes back to Mullenbach et al. (2018). None of these ideas has been applied to POLAR.

3 Task and Data

Inputs and outputs. The input is one social-media text, tokenised with the DeBERTa-v3 tokenizer and truncated to 256 tokens. Every model outputs a DET decision $\hat{d} \in \{0, 1\}$ and five TYPE decisions $\hat{t} \in \{0, 1\}^5$, together with the underlying probabilities.

Data. We use the official English data: 3,222 training, 160 development and 1,452 test texts (Naseem et al., 2026). Table 1 summarises the training split. The gold labels satisfy the hierarchy exactly in all three files. *Other* co-occurs with named dimensions in 121 training texts, so we model it as an ordinary fifth label rather than an exclusive residual. The supplied test file contains label columns whose provenance is unconfirmed. Until that is resolved, we treat it as strictly held out and do not use it. The 160-text development set

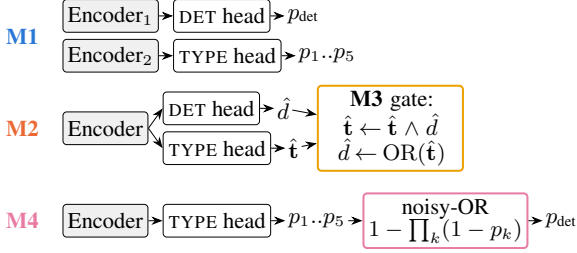


Figure 1: The four conditions. All share DeBERTa-v3-base, mean pooling and linear heads. M1 trains two separate encoders. M2 shares one encoder between an unconstrained DET head and TYPE head. M3 is not trained: it rewrites M2’s hard predictions after decoding. M4-core has no DET head; detection is a deterministic function of the five dimension probabilities, so the hierarchy holds by construction and the DET loss trains the TYPE head.

has single-digit positive counts for the rare dimensions, so it cannot support ablation claims (Bokaei et al., 2026). All conclusions therefore rest on out-of-fold predictions over the 3,222 training texts.

4 Models

All conditions share **DeBERTa-v3-base** (He et al., 2023), which has 184M parameters including embeddings. Each uses the attention-masked mean of the final hidden states, dropout 0.1 and linear heads (Figure 1). Let z_k be the logit of dimension k , $p_k = \sigma(z_k)$, and $\mathcal{P}$ the gold-polarized examples in a batch. The TYPE loss is masked to $\mathcal{P}$ in every trained model, because TYPE labels are undefined elsewhere (Mohammad, 2026). It uses per-label positive weights $w_k = \#\text{neg}_k / \#\text{pos}_k$, computed within $\mathcal{P}$.

M1 (independent). Two separately fine-tuned encoders, as in Guo and Chang (2026): M1-DET with binary cross-entropy (BCE), and M1-TYPE with masked weighted BCE. Their predictions are joined without reconciliation. M1 has twice the parameters (368M).

M2 (shared multi-task). One encoder with a DET head and a TYPE head:

$$\mathcal{L} = \text{BCE}(y_{\text{det}}, p_{\text{det}}) + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \mathcal{L}_{\text{type}}^{(i)}, \quad (1)$$

with $\lambda = 1$. Nothing ties the heads together, so M2 can violate the hierarchy in either direction.

M3 (post-hoc reconciliation). M3 is derived from M2’s thresholded predictions $\hat{d}$, $\hat{t}_k$, reusing

M2’s frozen thresholds. This is our reimplementation of the deterministic constraint of Maharjan et al. (2026). The *one-way* variant sets $\hat{t}_k \leftarrow \hat{t}_k \wedge \hat{d}$, which removes non-polarized-with-type violations only. The *symmetric* variant then sets $\hat{d} \leftarrow \bigvee_k \hat{t}_k$, which also removes polarized-without-type violations, so $\text{LVR} = 0$ is guaranteed. We keep the raw and reconciled predictions to audit what gating removes.

M4-core (structured noisy-OR). The same encoder and TYPE head as M2, with *no* DET head:

$$p_{\text{det}} = 1 - \prod_{k=1}^5 (1 - p_k), \quad (2)$$

trained with the loss of Eq. 1. We compute Eq. 2 in log space: with $s = \sum_k \text{softplus}(z_k)$, the logit is $\log(e^s - 1)$, evaluated with two numerically stable branches. On a non-polarized text the DET loss pushes *every* z_k down, which gives the dimension detectors supervision exactly where masking removes it. At inference, $\hat{t}_k = \mathbf{1}[p_k \geq \tau_k]$ and $\hat{d} = \bigvee_k \hat{t}_k$, so binary $\text{LVR} = 0$ holds by construction. M4-core omits the label-aware attention of our proposal on purpose. The mid-submission comparison changes one mechanism at a time, so any M3-versus-M4 difference can be attributed to the hierarchy rather than to the head architecture.

5 Experimental Protocol

Cross-validation. We create one fixed five-fold split with iterative multilabel stratification over DET and the five TYPE labels (Sechidis et al., 2011), using seed 2026. The folds hold 644–645 texts each, with rare labels balanced (14–15 Gender, 22–23 Religious per fold). The split is reused for every model and every seed (13, 21, 42, 87, 100). Within each outer split, 10% of the four training folds (258 texts) forms an inner validation set, stratified the same way. Early stopping, checkpoint selection and all thresholds use only this inner set. The outer fold is predicted exactly once, with the selected state and thresholds frozen. This yields 4 trained conditions $\times$ 5 folds $\times$ 5 seeds = **100 training runs**. M1 is joined, and both M3 variants are derived, for all 25 fold–seed pairs.

Training. AdamW (Loshchilov and Hutter, 2019), learning rate 2×10^{-5} , weight decay 0.01, batch size 8 with 2 accumulation steps, linear schedule with 10% warm-up, gradient clipping at 1.0, FP16 mixed precision, at most 8 epochs,

early-stopping patience 2. The checkpoint is selected on inner DET macro-F1 for M1-DET and on inner TYPE macro-F1 otherwise.

Thresholds. All thresholds are searched over $\{0.20, 0.25, \dots, 0.80\}$. Each TYPE threshold τ_k maximises that label’s F1 on the inner-validation polarized texts. M1-DET and M2 select a DET threshold that maximises DET macro-F1. M3 inherits M2’s thresholds, and M4-core has no DET threshold (Eq. 2 plus OR decoding).

Metrics. DET macro-F1 averages F1 over both classes. TYPE macro-P/R/F1 average over the five labels and, as frozen in our plan, are computed on *gold-polarized* texts. We additionally report TYPE on *all* texts as a pre-declared secondary view (§6.5). The symmetric Logical Violation Rate, extended from Maharjan et al. (2026) to both directions, is

$$\text{LVR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{d}^{(i)} \neq \bigvee_k \hat{t}_k^{(i)}], \quad (3)$$

split into LVR-A ($\hat{d} = 0$ with some $\hat{t}_k = 1$) and LVR-B ($\hat{d} = 1$ with no $\hat{t}_k = 1$). We also report label-level recall on multi-label and single-label polarized texts.

Aggregation and statistics. For each seed, the five outer folds are concatenated into one 3,222-text prediction set and scored once. Tables show the mean $\pm$ standard deviation of the five seed-level scores; we never treat the 25 fold-seed scores as independent. Paired comparisons use an example-level paired bootstrap (Koehn, 2004) with 2,000 resamples. Each resample applies the same multiset of texts to both models and all five seeds. The per-seed difference is averaged over seeds, and the 2.5/97.5 percentiles form the 95% CI. The six pre-declared primary tests (Table 3) are Holm-corrected (Holm, 1979). We also report how many of the five seeds favour each model. Secondary comparisons are exploratory and uncorrected.

Compute. All 100 runs ran on one NVIDIA RTX 4060 Laptop GPU (8 GB; peak memory 4.5 GB) with PyTorch 2.5.1 and Transformers 4.57.6. Each run took 1.9–2.9 minutes, 3.9 GPU-hours in total (Appendix B).

6 Results

Table 2 and Figure 2 give the overall picture. M1 is the strongest model on every primary metric, with M2 close behind on DET. The two consistent models, M3 and M4-core, are 0.04–0.09 TYPE macro-F1 below the inconsistent ones. The difference is driven by recall (0.47 vs. 0.60), not precision (0.47 vs. 0.44–0.51). M4-core has the smallest seed variance on TYPE (sd 0.007, against 0.013–0.022).

6.1 Hierarchy consistency

Figure 3 shows that the unconstrained models violate the hierarchy on **61.9%** (M1) and **61.6%** (M2) of all texts, nearly twice the highest rate Maharjan et al. (2026) report. Almost all of it is LVR-A. M2 attaches a dimension to every one of the 2,047 non-polarized texts, and M1 to 2,039 of them (Table 4).

This is a direct consequence of how the TYPE heads are trained. The masked loss means they never see a non-polarized text, the positive weights push their probabilities up, and their thresholds are tuned only on polarized texts. Nothing teaches them what non-polarized text looks like.

M3 (both variants) and M4-core reach exactly 0% in every run. That confirms the invariants of §4: symmetric M3 and M4-core have LVR = 0, and one-way M3 has LVR-A = 0. The interesting question is therefore not whether consistency is reached, but what it costs, which is RQ2.

6.2 RQ1: noisy-OR versus M1 and M2

Table 3 and Figure 4 give the verdict. M4-core is below M1 on DET (−0.014) and TYPE (−0.082) macro-F1, and below M2 on TYPE (−0.036); all three survive Holm correction, with 0/5 seeds in M4-core’s favour. Against M2 on DET, the difference (−0.008) is negative in every seed but not significant after correction ($p_{\text{Holm}} = 0.072$). **RQ1 is not supported** under the frozen protocol. Two mechanisms explain the outcome, and both matter for how far the result generalises.

The TYPE gap is the price of consistency. Because TYPE is scored only on gold-polarized texts, a model is never penalised for dimensions it predicts on non-polarized texts. M1 and M2 exploit this fully (§6.1). M4-core, like M3, can only be right on a polarized text if its detection also fires, so its TYPE recall is capped by its detection recall. The RQ1 TYPE comparison therefore pits a consistent model against inconsistent ones. The fair comparison at matched consistency is RQ2,

Model	Params	DET	TYPE (gold-polarized texts)			LVR (%)	
		macro-F1	macro-F1	macro-P	macro-R	A	total
M1 (independent)	368M	0.799 $\pm$ 0.007	0.543 $\pm$ 0.013	0.509 $\pm$ 0.024	0.600 $\pm$ 0.023	61.9	61.9
M2 (shared MTL)	184M	0.793 $\pm$ 0.003	0.497 $\pm$ 0.015	0.444 $\pm$ 0.018	0.595 $\pm$ 0.019	61.5	61.6
M3 one-way	184M	0.793 $\pm$ 0.003	0.453 $\pm$ 0.022	0.469 $\pm$ 0.033	0.468 $\pm$ 0.036	0.0	0.0
M3 symmetric	184M	0.793 $\pm$ 0.003	0.453 $\pm$ 0.022	0.469 $\pm$ 0.033	0.468 $\pm$ 0.036	0.0	0.0
M4-core (noisy-OR)	184M	0.784 $\pm$ 0.004	0.461 $\pm$ 0.007	0.465 $\pm$ 0.011	0.474 $\pm$ 0.022	0.0	0.0

Table 2: **Main results** (out-of-fold over 3,222 texts; mean $\pm$ sd over 5 seeds; best value per column in bold). TYPE is scored on gold-polarized texts, as frozen in the plan. LVR-B is 0.0% for all models: M3 one-way retains 0.4 violating texts per seed, and M2 essentially never predicts DET= 1 without any TYPE. M3’s two variants therefore have identical TYPE predictions and practically identical DET.

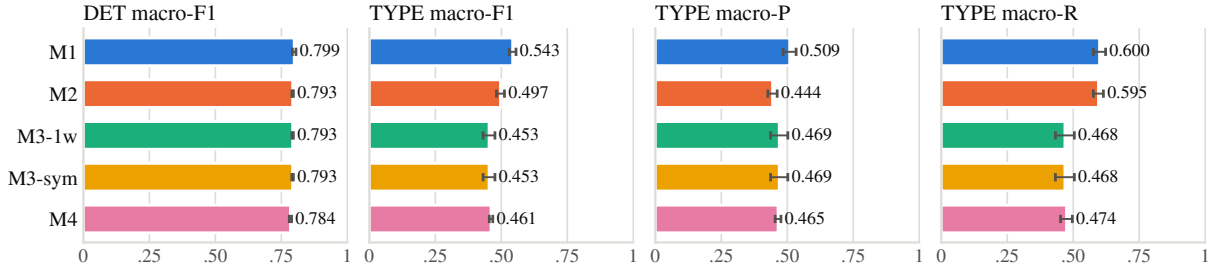


Figure 2: **Primary metrics by model.** Bars show the mean of the five seed-level out-of-fold scores, with whiskers for ± 1 sd. Read each panel from top to bottom as the model ladder: independent (M1), shared (M2), gated (M3) and structured (M4). DET is almost flat (0.784–0.799). On TYPE, the two consistent models (M3, M4) sit together, well below the two inconsistent ones (M1, M2), and the gap comes almost entirely from recall (right panel).

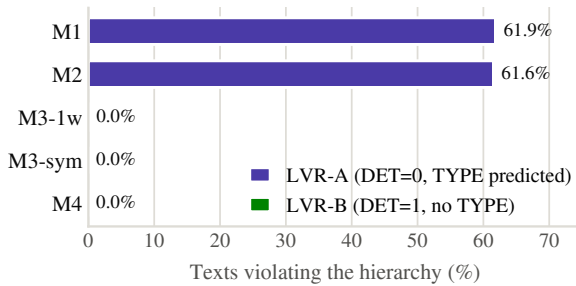


Figure 3: **Hierarchy violations.** Each bar is the share of all 3,222 texts whose hard predictions contradict the hierarchy, averaged over seeds. Purple is LVR-A (“not polarized” with a dimension); green is LVR-B (“polarized” with no dimension). M1 and M2 contradict themselves on 62% of texts, almost entirely in direction A. M3 and M4-core never do.

and when the same predictions are scored on all texts the sign reverses (§6.5). Restricted to polarized texts, M4-core is still *more precise* than M2 (+0.021 TYPE macro-P, CI [+0.001, +0.039], 4/5 seeds; Appendix Table 14).

The DET gap is a decoding artifact. M4-core over-detects (Table 4): 472 false positives per seed against 345 for M2, and 199 misses against 281. Its DET recall is 0.831 against 0.761 for M2 (+0.070, CI [+0.059, +0.081]), while precision

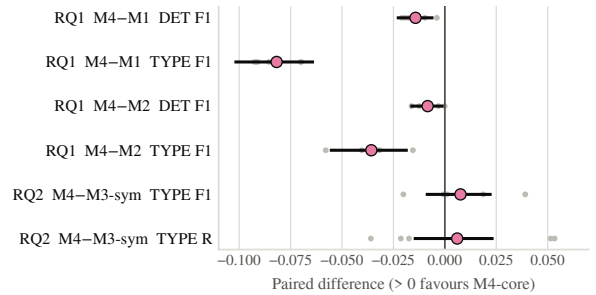


Figure 4: **Primary comparisons as paired differences.** Pink points are the seed-averaged difference (M4-core minus the baseline), black bars the 95% bootstrap CI, and grey dots the five per-seed differences. A bar that crosses the vertical zero line is not significant. All four RQ1 intervals lie left of zero (M4-core worse). Both RQ2 intervals straddle zero (no detectable difference), with seeds scattered on both sides.

falls from 0.722 to 0.674.

The decoding rule explains this. M4-core’s DET decision is the OR of five thresholds, each tuned for *its own label’s* F1 on polarized texts only, so no tuning objective ever sees the cost of a false alarm on a non-polarized text. The Political threshold, which effectively decides detection, is driven to the bottom of the grid: mean 0.22, and exactly 0.20 in 19 of 25 runs (Appendix Table 11). M1

RQ	Comparison & metric	A	B	Δ (A–B)	95% CI	p_{Holm}	Seeds
RQ1	M4–M1: DET macro-F1	0.784	0.799	−0.014	[−0.023, −0.006]	<0.001	0/5
RQ1	M4–M1: TYPE macro-F1	0.461	0.543	−0.082	[−0.102, −0.064]	<0.001	0/5
RQ1	M4–M2: DET macro-F1	0.784	0.793	−0.008	[−0.016, −0.001]	0.072	0/5
RQ1	M4–M2: TYPE macro-F1	0.461	0.497	−0.036	[−0.055, −0.019]	0.004	0/5
RQ2	M4–M3-sym: TYPE macro-F1	0.461	0.453	+0.008	[−0.009, +0.022]	0.732	3/5
RQ2	M4–M3-sym: TYPE macro-R	0.474	0.468	+0.006	[−0.014, +0.023]	0.732	2/5

Table 3: **Pre-declared primary comparisons.** A and B are seed-averaged scores and Δ is A–B. The CI is a 95% paired bootstrap interval over texts (2,000 resamples, averaged over seeds); p is two-sided and Holm-corrected over the six tests. “Seeds” counts the seeds (out of 5) whose out-of-fold score favours A.

Model	FP	FN	Pred. pos.	Non-pol. w/ TYPE
M1 (independent)	325	280	1,220	2,039
M2 (shared MTL)	345	281	1,239	2,047
M3 one-way	345	281	1,239	345
M3 symmetric	345	281	1,239	345
M4-core (noisy-OR)	472	199	1,448	472

Table 4: **DET error profile** (per seed, over 3,222 texts). FP: non-polarized texts predicted polarized. FN: polarized texts missed. Last column: non-polarized texts receiving at least one TYPE label. M4-core trades 82 fewer misses for 127 more false alarms than M2.

and M2 instead tune a dedicated DET threshold for DET macro-F1, and M1-DET also selects its checkpoint on DET.

The DET comparison is therefore not decoding-matched. We report this as a property of the frozen protocol rather than adjusting it after seeing results.

Sharing alone also hurts. M2 is below M1 by −0.046 TYPE F1 (CI [−0.060, −0.033]) and −0.006 DET F1. This is consistent with the negative transfer on sparse labels reported by [Guo and Chang \(2026\)](#). However, M1 uses twice the parameters and a DET-selected checkpoint.

6.3 RQ2: post-hoc gating versus training-time structure

With both M3 and M4-core at LVR = 0, the primary comparisons are statistically null (Table 3). M4-core’s TYPE F1 is +0.008 higher (CI [−0.009, +0.022], 3/5 seeds) and its TYPE recall +0.006 (CI [−0.014, +0.023], 2/5 seeds), with $p_{\text{Holm}} = 0.73$ for both. **RQ2 is inconclusive.** The data support neither “structure preserves more recall than gating” nor the reverse, and we report it as such.

Three secondary results qualify this:

1. On the 426 **multi-label** texts, M4-core recovers more gold labels than M3: label recall 0.610 vs. 0.585, +0.025 (CI [+0.009,

Label	n	Δ recall [95% CI]	Δ F1 [95% CI]
Political	1,150	+0.067 [+0.055, +0.078]	+0.040 [+0.033, +0.048]
Racial/ethnic	281	−0.011 [−0.041, +0.019]	−0.026 [−0.051, −0.003]
Religious	112	−0.016 [−0.059, +0.026]	−0.006 [−0.041, +0.030]
Gender/sexual	72	+0.064 [+0.009, +0.116]	+0.039 [−0.014, +0.089]
Other	126	−0.073 [−0.113, −0.034]	−0.009 [−0.039, +0.019]

Table 5: **RQ2 per label:** M4-core minus M3-symmetric on gold-polarized texts, with 95% paired bootstrap CIs. Bold marks intervals that exclude zero. The macro average hides opposing per-label effects.

+0.041], 4/5 seeds; Table 7). This is the direction [Maharjan et al. \(2026\)](#) anticipate (“gating limits multi-label recall”), but it is an exploratory comparison.

2. M4-core is **more stable**: its TYPE F1 seed sd is 0.007, against 0.022 for M3.
3. M4-core pays −0.008 DET macro-F1 (CI [−0.016, −0.001]) for its broader detection.

Per-label effects cancel in the macro average. Table 5 and Figure 5 show opposite movements underneath the null macro result.

- **Gains.** M4-core gains recall on Political (+0.067) and on the rarest label, Gender/sexual (+0.064, CI [+0.009, +0.116]).
- **Losses.** It loses recall on *Other* (−0.073) and F1 on Racial/ethnic (−0.026).

One plausible reading is that *Other*, the most heterogeneous label, absorbs the strongest push-down from the noisy-OR DET loss on non-polarized text. That motivates the separate treatment of *Other* proposed for C3.

6.4 Gating audit: what M3 removes

Figure 6 and Table 6 make C2’s mechanism concrete. Per seed, gating removes 4,379 of the 6,647 dimension labels M2 predicts, and 87.7% of the removals are false alarms on non-polarized texts.

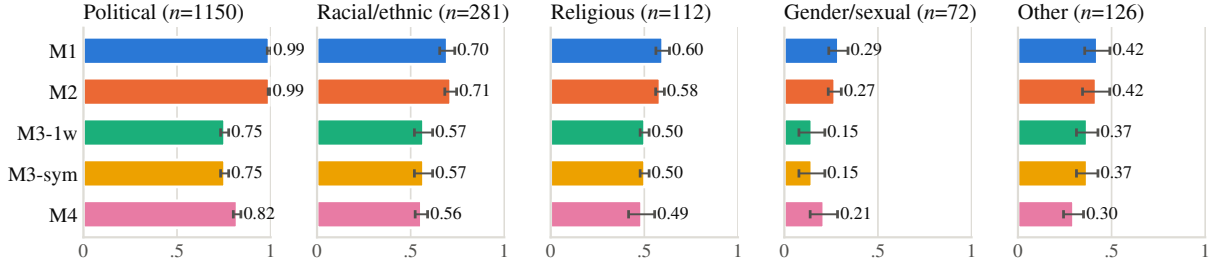


Figure 5: **Per-label TYPE recall on gold-polarized texts** (mean $\pm$ sd over seeds; n = gold support). Comparing the yellow (M3) and pink (M4-core) bars isolates RQ2 per label. M4-core recovers recall on Political, where its higher detection recall matters most, and on the rarest label, Gender/sexual. It loses recall on *Other*. The gap between the top two bars and the rest is the recall that consistency costs on each label.

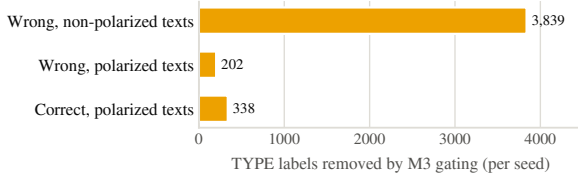


Figure 6: **What gating removes.** The three bars partition the TYPE labels that M3 deletes from M2’s predictions, per seed. The long top bar is gating working as intended: wrong labels on non-polarized texts. The bottom bar is its cost: correct labels deleted because M2’s detection head missed a polarized text.

Quantity (per seed)	Mean $\pm$ sd	Share
TYPE labels predicted by M2	6,647 $\pm$ 775	
Labels removed by gating	4,379 $\pm$ 767	
wrong, on non-polarized texts	3,839 $\pm$ 726	87.7%
wrong, on polarized texts	202 $\pm$ 51	4.6%
correct , on polarized texts	338 $\pm$ 29	7.7%
Polarized texts gated (M2 DET misses)	281 $\pm$ 25	
with ≥ 1 correct M2 TYPE label	277 $\pm$ 26	98.7%
DET decisions changed by OR step	0 $\pm$ 1	

Table 6: **M2→M3 gating audit** (per seed over 3,222 texts; mean $\pm$ sd over seeds). Shares are relative to labels removed, or to polarized texts gated for the last share row. The 338 correct labels removed are 22.9% of the 1,479 correct TYPE labels M2 predicts.

That is gating doing its job. It also removes **338 correct labels**, 22.9% of everything M2 gets right.

The reason is error propagation down the hierarchy. M2’s DET head misses 281 polarized texts per seed, and on **98.7%** of them (277) M2 had already predicted at least one correct dimension. Gating discards those labels because the parent decision was wrong. The symmetric OR step almost never fires (0.4 DET changes per seed).

On the primary metric, this costs M3 -0.127 TYPE recall (CI $[-0.145, -0.110]$) and -0.043 TYPE F1 relative to M2, and -0.161 recall on multi-label texts. It gains only $+0.025$ precision.

The loss is much larger than the $0.511 \rightarrow 0.506$ drop of Maharjan et al. (2026); their scoring scope may differ from ours (§6.5).

6.5 Sensitivity to the TYPE scoring scope

Scoring TYPE on all texts counts every dimension predicted for a non-polarized text as a false positive. This view is pre-declared, and it matches the leaderboard setting if the official scorer evaluates Subtask 2 over all texts, which we have not yet verified.

Under it the ranking *reverses* (Table 7). M4-core (0.396) $>$ M3 (0.382) $>$ M1 (0.299) $>$ M2 (0.255). M4-core exceeds M1 by $+0.097$ (CI $[+0.078, +0.114]$) and M2 by $+0.141$ (CI $[+0.121, +0.158]$), in 5/5 seeds each. Against M3 it gains $+0.014$ (CI $[-0.001, +0.026]$, 4/5 seeds), while M3 itself gains $+0.127$ over M2.

The RQ1 conclusion is therefore *conditional on the scoring scope*. The consistent models lose to the inconsistent ones only when the inconsistency is not penalised. Appendix Figure 8 plots these metrics.

7 Discussion

What the evidence supports. The diagnostic part of C2 holds on its own. Unconstrained heads trained with a masked TYPE loss are almost maximally inconsistent (62% of texts, essentially every non-polarized one). Post-hoc gating fixes this at a quantifiable price: 23% of correct labels, removed almost exclusively through detection misses. Training-time structure reaches the same consistency with equal macro accuracy, better multi-label recall and lower variance. The evidence does *not* support the claims that noisy-OR improves macro-F1 over independent or multi-task models (RQ1), or that structure preserves

Model	DET (positive class)		TYPE on all 3,222 texts			Label recall	
	P	R	macro-F1	macro-P	macro-R	multi-label	single-label
M1 (independent)	0.733 $\pm$ 0.002	0.761 $\pm$ 0.026	0.299 $\pm$ 0.013	0.203 $\pm$ 0.014	0.600 $\pm$ 0.023	0.743 $\pm$ 0.020	0.987 $\pm$ 0.002
M2 (shared MTL)	0.722 $\pm$ 0.009	0.761 $\pm$ 0.021	0.255 $\pm$ 0.029	0.169 $\pm$ 0.023	0.595 $\pm$ 0.019	0.746 $\pm$ 0.017	0.987 $\pm$ 0.004
M3 one-way	0.722 $\pm$ 0.009	0.761 $\pm$ 0.021	0.382 $\pm$ 0.017	0.339 $\pm$ 0.019	0.468 $\pm$ 0.036	0.585 $\pm$ 0.033	0.748 $\pm$ 0.027
M3 symmetric	0.722 $\pm$ 0.009	0.761 $\pm$ 0.021	0.382 $\pm$ 0.017	0.339 $\pm$ 0.019	0.468 $\pm$ 0.036	0.585 $\pm$ 0.033	0.748 $\pm$ 0.027
M4-core (noisy-OR)	0.674 $\pm$ 0.008	0.831 $\pm$ 0.015	0.396 $\pm$ 0.008	0.348 $\pm$ 0.008	0.474 $\pm$ 0.022	0.610 $\pm$ 0.018	0.805 $\pm$ 0.010

Table 7: **Secondary metrics.** DET precision and recall for the polarized class; TYPE scored on *all* 3,222 texts (non-polarized texts count as all-negative); label-level recall on the 426 multi-label and 749 single-label polarized texts. Best per column in bold.

Dates	Work package
Oct 2–6	Verify the official Subtask 2 scorer; DET-aware threshold search for M4-core (joint objective on all inner-validation texts, grid below 0.20); re-run M4-core (25 runs).
Oct 7–14	C3/RQ3: label-aware cross-attention heads and separate handling of <i>Other</i> ; ablations; single- vs. multi-label analysis.
Oct 15–20	Reference baselines from the proposal (TF-IDF+LR, BERT, RoBERTa); uncertainty-weighted λ (Kendall et al., 2018).
Oct 21–25	C4/RQ4: performance versus parameter count against the 12B–27B leaderboard systems; final fit with checkpoint retention; dev evaluation (and test, once provenance is confirmed).
Oct 26–31	Error analysis of <i>Other</i> and false-positive boundary cases; final report and code release.

Table 8: Plan to the final submission (October 31).

more TYPE recall than gating in the macro average (RQ2). Following our plan, we do not reframe either question after seeing the results.

Why the trade-off did not disappear. Our proposal hypothesised that M4 avoids gating’s recall loss because it “discards nothing”. The audit shows why that is incomplete: under *any* consistent decoder, a polarized text whose detection fails yields no dimension, so TYPE recall on polarized texts is bounded by DET recall. M4-core raises DET recall (0.831 vs. 0.761), which is where its Political and Gender gains come from, but pays in DET precision because its per-label thresholds never see non-polarized text. The binding constraint is the *decoder*, not the training objective, which makes a DET-aware threshold search the most informative next experiment.

8 Remaining Work and Timeline

Table 8 lists the remaining work. Three changes to the proposal timeline follow from the mid-

submission results.

1. M3 is already complete. It was a re-implementation of the Maharjan et al. (2026) rule rather than their released code, so its slot is reassigned to fixing M4-core’s decoder.
2. The classical and BERT/RoBERTa baselines were deferred so that the controlled M1–M4 ladder could be completed first.
3. Any revised decoding rule is a *protocol extension*, reported separately from the frozen RQ1/RQ2 answers given here.

9 Conclusion

We built and ran a controlled, fully paired comparison of four ways to handle the POLAR detection/dimension hierarchy: 100 training runs, 25 derived conditions per post-processed model, and bootstrap CIs with multiplicity control.

- Unconstrained models contradict the hierarchy on most texts.
- Post-hoc gating removes the contradictions but also nearly a quarter of the correct dimension labels.
- A noisy-OR model reaches the same consistency with statistically equal macro accuracy, better multi-label recall and lower variance, but it does not beat the unconstrained baselines on the frozen metric.

Whether structure can *eliminate* the consistency–recall trade-off now hinges on its decoder, which the next stage addresses.

Limitations

- **Scoring scope.** TYPE is scored on gold-polarized texts, as frozen before the experiments. This favours unconstrained models.

We have not yet confirmed how the official scorer treats non-polarized texts.

- **Decoding not matched.** M1 and M2 tune a DET threshold for DET; M4-core cannot. Checkpoint selection also differs (M1-DET on DET, others on TYPE). The Political threshold of M4-core hits the grid floor in 19/25 runs.
- **Parameter mismatch.** M1 uses 368M parameters against 184M for the other conditions.
- **Uncertainty.** Bootstrap CIs capture sampling variation over texts. Training randomness is reflected only through five seeds. Only the six primary tests are multiplicity-corrected.
- **Scope.** The results cover English cross-validation only. Weights were not retained in the sweep, so the development and test sets are not yet scored, and absolute numbers are not leaderboard-comparable.
- **Training budget.** Training is capped at 8 epochs, and 3 of 100 runs selected the final epoch.
- **Hardware.** The runs used one consumer GPU rather than the planned cluster. Every condition used the same machine, so the comparison is internally consistent.

Ethical Considerations

We use the released POLAR data without collecting new data. The texts contain hostile and stereotyping content about protected groups. Polarization classifiers can be misused for surveillance or for silencing legitimate political speech, and their errors fall unevenly on the rare dimensions, where our models are weakest (Gender/sexual F1 ≤ 0.27). We report per-label results so that these weaknesses stay visible.

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A Implementation and Reproducibility

Setting	Value
Encoder	microsoft/deberta-v3-base, revision de19fe7 (safetensors; tensor-identical to main 8ccc9b6)
Pooling / dropout / max length	attention-masked mean / 0.1 / 256 tokens
Optimiser	AdamW, lr 2×10^{-5} , weight decay 0.01, linear schedule, 10% warm-up, clip 1.0
Batch	8 per step, 2 accumulation steps (effective 16); eval batch 16
Epochs	≤ 8 , early-stopping patience 2 on the selection metric
Loss	BCE (DET) + λ masked weighted BCE (TYPE), $\lambda = 1$, $w_k = \#neg_k / \#pos_k$ within polarized texts
Precision	FP16 autocast with FP32 master weights and dynamic loss scaling
Folds	5, iterative multilabel stratification over DET+5 TYPE, seed 2026
Inner validation	10% of the four training folds, multilabel-stratified, seeded by the run seed
Seeds	13, 21, 42, 87, 100
Threshold grid	0.20–0.80, step 0.05; per-label TYPE thresholds; DET threshold for M1/M2
Bootstrap	2,000 paired resamples of texts, seed 2026; percentile CIs; Holm over 6 primary tests
Software / hardware	PyTorch 2.5.1 (CUDA 12.1), Transformers 4.57.6, scikit-learn 1.6.1; NVIDIA RTX 4060 Laptop GPU
Tracking	every run, with its prediction file, logged to Weights & Biases

Table 9: Hyperparameters and implementation details shared by all trained conditions.

Pipeline. `create_folds.py` writes the fixed fold file. `run_local_sweep.sh` trains the 100 runs, resumably and seed-major. `combine_m1.py` and `reconcile_m3.py` derive M1 and both M3 variants per fold–seed. `analyze_mid_submission.py` computes every metric, comparison and audit from the saved prediction files; its vectorised metrics are asserted equal to the project’s reference implementation. `make_report_assets.py` produces every table and figure in this report. All numbers are therefore regenerated from saved out-of-fold predictions, never from dashboard summaries.

Checks. Each of the 20 model–seed prediction sets covers exactly the 3,222 texts once, with matching fold labels. Every run logged one completion event with no NaNs. The hierarchy invariants of §6.1 hold in every run. Two defects were fixed before the sweep, each covered by a unit test (8/8 pass): the M1-DET forward pass rejected the TYPE labels present in shared batches, and the noisy-OR logit had a NaN-gradient branch when all five logits were ≤ -19 .

B Training Statistics and Thresholds

Run type	Best epoch	Range	At cap	Epochs run	Min/run
M1-DET	2.8	1–5	0	4.8	1.9
M1-TYPE	4.1	3–6	0	6.1	2.4
M2	3.9	2–8	2	5.8	2.3
M4-core	5.8	4–8	1	7.5	2.9
Total: 100 runs, 3.9 GPU-hours					

Table 10: Training statistics over the 25 runs of each trained condition. “At cap” counts runs whose best epoch was the last allowed (8). M4-core converges more slowly than M2: its DET gradient reaches the encoder only through the product of five sigmoids.

Run type	DET	Political	Racial	Religious	Gender	Other
M1-DET	0.47 (3)	–	–	–	–	–
M1-TYPE	–	0.31 (7)	0.56 (2)	0.63 (6)	0.43 (6)	0.40 (6)
M2	0.49 (6)	0.27 (4)	0.48 (0)	0.66 (7)	0.57 (1)	0.45 (4)
M4-core	–	0.22 (19)	0.50 (2)	0.47 (1)	0.50 (2)	0.47 (7)

Table 11: Mean selected thresholds over 25 runs, with the number of runs at the grid edge (0.20 or 0.80) in parentheses. M4-core’s Political threshold is at the floor in 19/25 runs, which drives its DET over-prediction (§6.2).

C Full Per-Label Results

	Model	Political ($n=1,150$)	Racial/ethnic ($n=281$)	Religious ($n=112$)	Gender/sexual ($n=72$)	Other ($n=126$)
F1	M1	0.985 ± 0.005	0.623 ± 0.024	0.548 ± 0.019	0.266 ± 0.041	0.293 ± 0.019
	M2	0.986 ± 0.003	0.583 ± 0.021	0.477 ± 0.035	0.200 ± 0.060	0.238 ± 0.019
	M3-1w	0.852 ± 0.015	0.547 ± 0.020	0.456 ± 0.030	0.175 ± 0.081	0.237 ± 0.022
	M3-sym	0.852 ± 0.015	0.547 ± 0.020	0.456 ± 0.030	0.175 ± 0.081	0.237 ± 0.022
	M4	0.892 ± 0.012	0.521 ± 0.013	0.450 ± 0.017	0.214 ± 0.035	0.228 ± 0.007
P	M1	0.979 ± 0.001	0.565 ± 0.033	0.508 ± 0.041	0.265 ± 0.078	0.227 ± 0.022
	M2	0.979 ± 0.001	0.494 ± 0.034	0.409 ± 0.069	0.169 ± 0.075	0.169 ± 0.023
	M3-1w	0.979 ± 0.003	0.532 ± 0.044	0.424 ± 0.067	0.235 ± 0.121	0.178 ± 0.030
	M3-sym	0.979 ± 0.003	0.532 ± 0.044	0.424 ± 0.067	0.235 ± 0.121	0.178 ± 0.030
	M4	0.978 ± 0.002	0.490 ± 0.012	0.425 ± 0.022	0.241 ± 0.040	0.191 ± 0.028
R	M1	0.991 ± 0.009	0.695 ± 0.040	0.598 ± 0.036	0.289 ± 0.051	0.424 ± 0.067
	M2	0.992 ± 0.005	0.714 ± 0.032	0.584 ± 0.023	0.269 ± 0.035	0.417 ± 0.073
	M3-1w	0.754 ± 0.022	0.569 ± 0.049	0.502 ± 0.024	0.147 ± 0.069	0.370 ± 0.057
	M3-sym	0.754 ± 0.022	0.569 ± 0.049	0.502 ± 0.024	0.147 ± 0.069	0.370 ± 0.057
	M4	0.821 ± 0.020	0.557 ± 0.033	0.486 ± 0.070	0.211 ± 0.073	0.297 ± 0.053

Table 12: Per-label F1, precision (P) and recall (R) on gold-polarized texts (mean $\pm$ sd over seeds). M3-1w and M3-sym coincide because their TYPE predictions are identical.

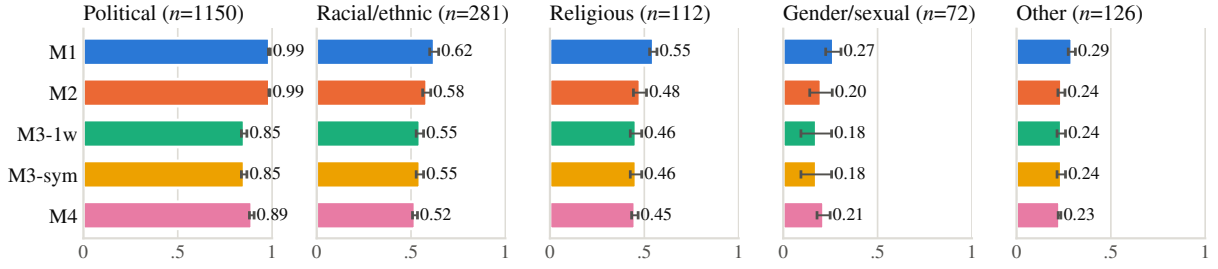


Figure 7: Per-label TYPE F1 on gold-polarized texts (mean $\pm$ sd over seeds). Gender/sexual and *Other* are hard for every model ($F1 \leq 0.29$) and show the largest seed variance; their thresholds are tuned on only about 6 and 10 inner-validation positives. M4-core has the smallest variance on the rare labels.

Comparison	Label	Δ precision	Δ recall	Δ F1
M4–M3-sym	Political	–0.001	+0.067	+0.040
M4–M3-sym	Racial/ethnic	–0.042	–0.011	–0.026
M4–M3-sym	Religious	+0.001	–0.016	–0.006
M4–M3-sym	Gender/sexual	+0.006	+0.064	+0.039
M4–M3-sym	Other	+0.013	–0.073	–0.009
M4–M2	Political	–0.001	–0.171	–0.093
M4–M2	Racial/ethnic	–0.004	–0.157	–0.062
M4–M2	Religious	+0.016	–0.098	–0.027
M4–M2	Gender/sexual	+0.072	–0.058	+0.014
M4–M2	Other	+0.022	–0.121	–0.010
M4–M1	Political	–0.001	–0.171	–0.093
M4–M1	Racial/ethnic	–0.075	–0.138	–0.102
M4–M1	Religious	–0.083	–0.112	–0.098
M4–M1	Gender/sexual	–0.024	–0.078	–0.052
M4–M1	Other	–0.036	–0.127	–0.065

Table 13: Per-label paired differences of M4-core against each model (gold-polarized texts). Bold: 95% bootstrap CI excludes zero. Against M1 and M2, M4-core loses recall on every label but keeps precision; against M3, the effects are mixed.

D Secondary Comparisons

RQ	Comparison & metric	A	B	Δ (A–B)	95% CI	<i>p</i>	Seeds
RQ1	M4–M1: TYPE macro-P	0.465	0.509	−0.044	[−0.064, −0.026]	<0.001	0/5
RQ1	M4–M1: TYPE macro-R	0.474	0.600	−0.125	[−0.151, −0.101]	<0.001	0/5
RQ1	M4–M2: TYPE macro-P	0.465	0.444	+0.021	[+0.001, +0.039]	0.037	4/5
RQ1	M4–M2: TYPE macro-R	0.474	0.595	−0.121	[−0.148, −0.098]	<0.001	0/5
RQ1	M4–M1: DET precision	0.674	0.733	−0.059	[−0.069, −0.048]	<0.001	0/5
RQ1	M4–M1: DET recall	0.831	0.761	+0.069	[+0.057, +0.082]	<0.001	5/5
RQ1	M4–M2: DET precision	0.674	0.722	−0.048	[−0.057, −0.038]	<0.001	0/5
RQ1	M4–M2: DET recall	0.831	0.761	+0.070	[+0.059, +0.081]	<0.001	5/5
RQ1	M4–M1: TYPE macro-F1 (all texts)	0.396	0.299	+0.097	[+0.078, +0.114]	<0.001	5/5
RQ1	M4–M2: TYPE macro-F1 (all texts)	0.396	0.255	+0.141	[+0.121, +0.158]	<0.001	5/5
RQ1	M2–M1: DET macro-F1	0.793	0.799	−0.006	[−0.011, −0.000]	0.039	1/5
RQ1	M2–M1: TYPE macro-F1	0.497	0.543	−0.046	[−0.060, −0.033]	<0.001	0/5
RQ2	M4–M3-sym: TYPE macro-P	0.465	0.469	−0.004	[−0.022, +0.012]	0.553	2/5
RQ2	M4–M3-sym: DET macro-F1	0.784	0.793	−0.008	[−0.016, −0.001]	0.026	0/5
RQ2	M4–M3-sym: Label recall (multi-label texts)	0.610	0.585	+0.025	[+0.009, +0.041]	0.001	4/5
RQ2	M4–M3-sym: TYPE macro-F1 (all texts)	0.396	0.382	+0.014	[−0.001, +0.026]	0.073	4/5
RQ2	M3-sym–M2: TYPE macro-F1	0.453	0.497	−0.043	[−0.056, −0.032]	<0.001	0/5
RQ2	M3-sym–M2: TYPE macro-R	0.468	0.595	−0.127	[−0.145, −0.110]	<0.001	0/5
RQ2	M3-sym–M2: TYPE macro-P	0.469	0.444	+0.025	[+0.013, +0.038]	<0.001	5/5
RQ2	M3-sym–M2: DET macro-F1	0.793	0.793	−0.000	[−0.000, +0.000]	0.242	0/5
RQ2	M3-sym–M2: TYPE macro-F1 (all texts)	0.382	0.255	+0.127	[+0.113, +0.142]	<0.001	5/5
RQ2	M3-sym–M2: Label recall (multi-label texts)	0.585	0.746	−0.161	[−0.184, −0.137]	<0.001	0/5

Table 14: Exploratory secondary comparisons (95% paired bootstrap CIs; *p* uncorrected). Rows include the effect of sharing (M2–M1) and the effect of gating (M3-sym–M2).

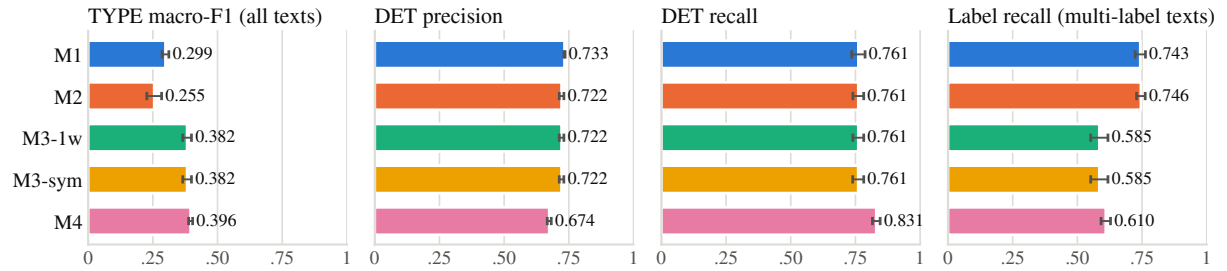


Figure 8: **Secondary metrics.** Left: TYPE macro-F1 on all texts, where the inconsistent models (M1, M2) fall below the consistent ones because their labels on non-polarized texts count as errors. Middle: DET precision and recall, showing M4-core’s shift toward recall. Right: label recall on multi-label texts, where gating (M3) loses most and M4-core recovers part of it.